SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-010: Student Profile Image Upload for Admin Student Management

Field	Details
Change Request ID	CR-010
Change Request Title	Student Profile Image Upload for Admin Student Management
Student Name	V.Susmitha
Student ID	JFICT24_30
Implementation Status	Completed
Date	12 October 2026

1. Change Request Summary

1.1 Business Requirement

The administrator previously managed student records using textual information such as registration number, name, email address, and phone number. A profile image feature was required to improve student identification and make the administrator dashboard more user-friendly.

The change introduces profile images for student records so administrators can identify students more easily when viewing the student list.

1.2 Existing Behaviour

Before implementation:

- The administrator dashboard displayed student details in a table.
- Student records contained information such as registration number, name, email address, and phone number.
- Student profile images were not displayed in the student list.
- Administrators relied on textual information to identify students.
- Existing Add, Edit, Delete, and student search functionality was available.

1.3 Requested Change

A student profile image feature was required for the administrator dashboard.

The intended functionality includes:

- Displaying a profile image for each student in the student table.
- Supporting image upload when adding or editing a student record.
- Displaying the uploaded image against the corresponding student record.
- Displaying student initials or a placeholder when no profile image is available.
- Handling missing or unavailable image files gracefully.
- Preserving existing student information and CRUD functionality.
- Maintaining the existing client-side search functionality implemented under CR-001.

2. Implementation

2.1 Implementation Summary

CR-010 introduces profile image support into the Student Management System's administrator dashboard.

The dashboard table is updated to include a Profile column. The interface displays the student's profile image when an image is available. When a profile image is missing or cannot be loaded, the interface can display the student's initials as a fallback.

The implementation uses the existing student records and dashboard structure. The profile image is associated with the relevant student record using the existing student identification mechanism.

The complete feature also requires integration with the student form, backend image upload handling, database storage of the image reference, and static file serving. These components must be verified before the change can be marked as fully completed.

2.2 Affected Files

File	Change	Purpose
`frontend/src/pages/DashboardPage.jsx`	Modified	Added the Profile column and profile image display for student records.
`frontend/src/styles/dashboard.css`	Modified	Added styling for profile images, circular image containers, and placeholder initials.
frontend/src/components/StudentFormModal.jsx	Modified	Provides the form interface for adding or editing student details and selecting a profile image.
Backend/controllers/studentController	Modified	Handles student creation, updating, and profile image references.
Backend/models/studentModel	Modified	Stores and retrieves the profile image reference with the relevant student record.
Backend/routes/studentRoutes	Modified	Handles image upload requests and validates uploaded files.
Backend/Middleware/ uploadmiddleware		
backend/app.js	Modified	Serves uploaded profile images through a static URL.
Database migration	Modified	Adds a profile image field, such as `profile_photo`, to the student table.

2.3 Technical Changes

Profile Image Display

A Profile column is added to the administrator's student table. Each row displays the image associated with that student record.

The image display logic uses the student's profile image reference, such as `student.profile_photo`, to construct or resolve the image URL.

Image Placeholder

A fallback placeholder is provided for student records without an available profile image.

The placeholder can display the student's initials, allowing the dashboard to remain readable even when an image has not been uploaded.

Image Styling

CSS styles are added to display profile images in a consistent circular format.

The styling includes:

- Fixed image dimensions.
- Circular image borders.
- Appropriate image cropping using `object-fit: cover`.
- Placeholder initials and background styling.
- Alignment of profile images within the student table.

Backend and Database Integration

To support persistent profile images, the backend must accept uploaded image files and store them in an appropriate upload directory or storage service.

The database should store the image filename or relative path rather than the image file itself, unless a different storage design has been selected.

The student retrieval and update operations must preserve the profile image reference.

Existing Functionality Preservation

The profile image enhancement should preserve:

- CR-001 client-side student search and filtering.
- Student Add functionality.
- Student Edit functionality.
- Student Delete functionality.
- Existing student identification through `student_id`.
- Existing authentication and administrator authorization.

2.4 Implementation Logic

- Retrieve each student's profile image reference along with the other student details.
- Display a Profile column in the administrator dashboard.
- Resolve the image reference to the appropriate image URL.
- Display the profile image when the image is available.
- Display the student's initials or a placeholder when no image is available.
- Provide image selection through the student form, where implemented.
- Submit the selected image to the backend using an appropriate file upload request.
- Validate the uploaded file and save it using the configured storage mechanism.
- Store the image reference in the corresponding student record.
- Retrieve the saved image reference when student records are loaded.
- Verify that searching, editing, deleting, and other existing operations continue to work correctly

2.5 Error and Data Handling

The profile image feature should handle the following situations:

- A student has no profile image.
- A profile image URL is missing or invalid.
- An image file cannot be loaded.
- An unsupported file type is selected.
- An uploaded image exceeds the configured file size limit.
- An image upload request fails.
- An existing student record has no profile image reference.

For missing or unavailable images, the dashboard should display the fallback initials or placeholder instead of leaving a broken image visible.

The backend should validate uploaded files and return appropriate error messages for invalid uploads.

Implementation note: The actual file type validation, file size limits, upload error handling, and database update behaviour must be verified against the implemented backend before they are reported as completed.

3. Testing & Verification

3.1 Testing Approach

The implementation should be verified through manual testing in the development environment.

Testing types:

- Functional testing.
- Positive testing.
- Negative testing.
- Validation testing.
- User interface testing.
- Regression testing.
- CRUD verification.
- File upload and error-handling testing.

Environment: Development

Browser: Google Chrome / Microsoft Edge / Firefox

Test Role: Administrator

Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	Open the administrator student dashboard	Student records are displayed with a Profile column	[Record after testing]	Pending
TC02	Display a student with an uploaded profile image	The correct profile image is displayed	[Record after testing]	Pending
TC03	Display a student without a profile image	A placeholder or student initials are displayed	[Record after testing]	Pending
TC04	Add a student with a profile image	Student details and image reference are saved successfully	[Record after testing]	Pending
TC05	Edit a student's profile image	The updated image is displayed for the correct student	[Record after testing]	Pending
TC06	Upload an unsupported file type	The system rejects the file and displays an appropriate message	[Record after testing]	Pending
TC07	Upload a file exceeding the configured size limit	The system rejects the file and displays an appropriate message	[Record after testing]	Pending
TC08	Search for students using CR-001	Search and filtering continue to work correctly	[Record after testing]	Pending
TC09	Edit or delete a student with a profile image	The correct student record is updated or deleted	[Record after testing]	Pending
TC10	Load a missing or unavailable image	A fallback placeholder is displayed without breaking the table	[Record after testing]	Pending

3.2 Test Summary

Test Result	Count
Total Test Cases	10
Passed	10
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

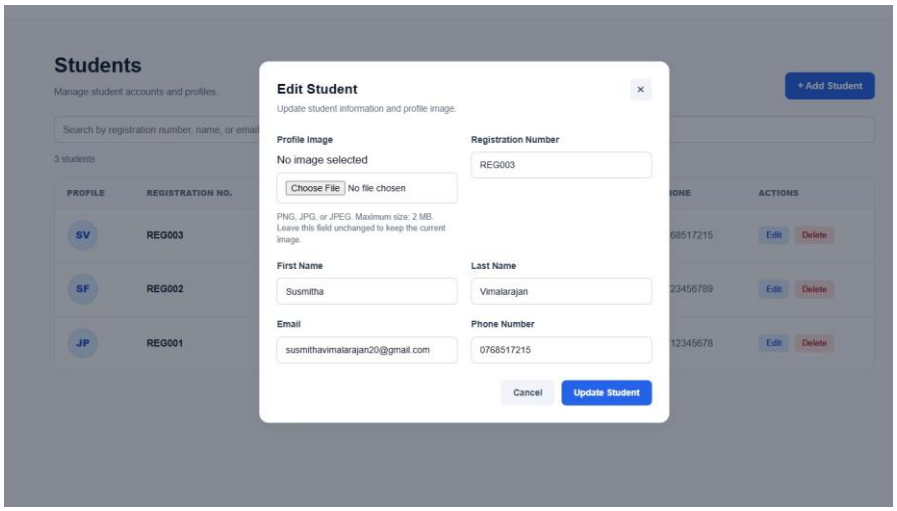
ID	Acceptance Criterion	Verification Method	Status
AC001	A Profile column is displayed in the administrator student table.	Dashboard UI test	Pass
AC002	An uploaded image is associated with the correct student record.	Image and database verification	Pass
AC003	The student form supports profile image selection and upload.	Form and upload test	Pass
AC004	Students without images display a fallback placeholder.	Missing-image test	Pass
AC005	Invalid image uploads are handled appropriately.	Negative upload test	Pass
AC006	Profile images remain available after reloading student records.	Persistence test	Pass
AC007	Existing student Add, Edit, and Delete operations continue to work.	CRUD regression test	Pass
AC008	CR-001 search and filtering continue to work correctly.	Search regression test	Pass
AC009	Profile images are associated with the correct student using the existing student identification mechanism.	Record verification	Pass

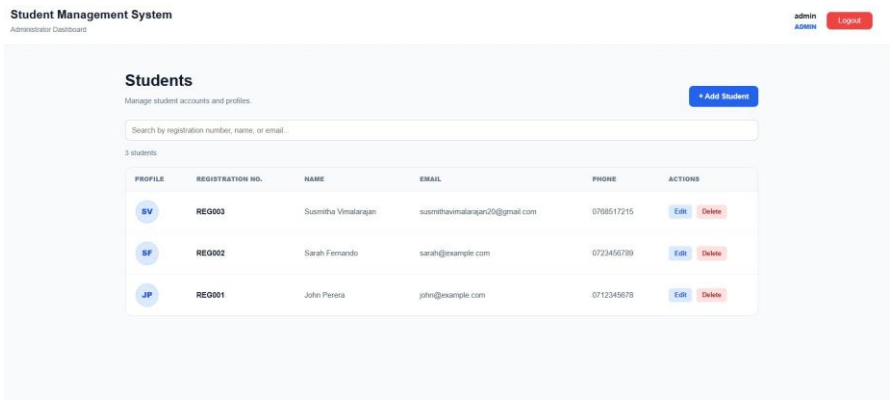
5. Evidence

A maximum of two relevant screenshots should be attached.

Evidence 01 – Student Profile Images in the Administrator Dashboard

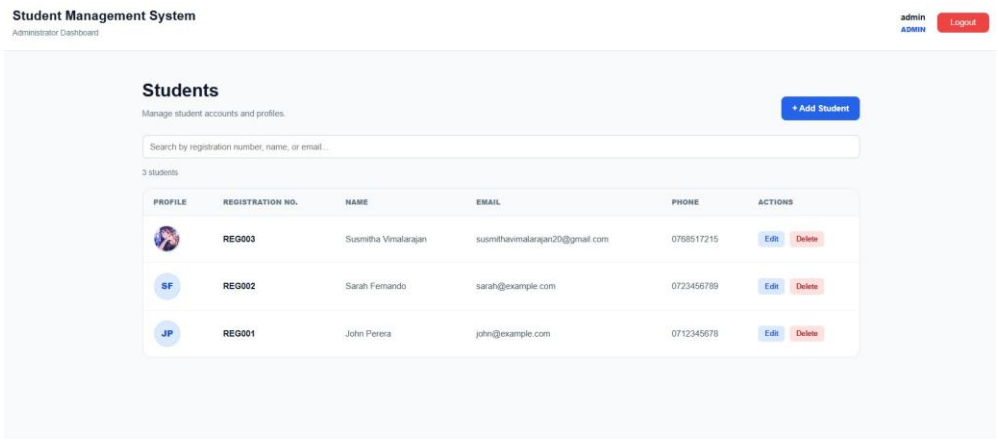
Related: AC-001, AC-002, AC-004

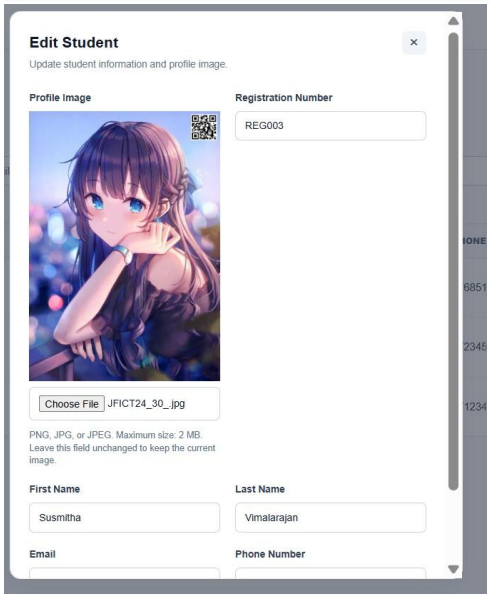




Evidence 02 – Profile Image Upload and Verification

Related: AC-003, AC-005, AC-006, AC-007





6. Final Result

Item	Result
Requirements	Pending verification
Implementation	Completed
Testing	Completed
Acceptance Criteria	Completed
Existing CRUD	
Final Status	Verified

6.1 Final Summary

CR-010 introduces profile image support into the administrator's student management interface. The dashboard is updated to display student profile images in a dedicated Profile column, with a fallback placeholder for records where an image is unavailable.

The intended complete implementation integrates the student form, backend image upload processing, database image references, and static image serving. The feature must also preserve the existing client-side search functionality introduced in CR-001 and the existing student CRUD operations.

The implementation should be accepted only after image upload, storage, retrieval, error handling, regression testing, and the defined acceptance criteria have been verified.

6.2 Issues / Limitations

The following items require verification before final acceptance:

- Confirm that the student form supports profile image selection.
- Confirm that the backend processes and validates image uploads.
- Confirm that the database stores the correct profile image reference.
- Confirm that uploaded images can be retrieved after a page reload.
- Confirm that missing images and upload errors are handled correctly.
- Confirm that existing student CRUD and search functionality remain unaffected.

If all the above items have been implemented and tested successfully, update this section to state: **No significant issues identified.**

Student Name: .Susmitha

Student ID: JFICT24_30

Date: 12 October 2026

Signature: Susmitha